ContractIQ – Sprint 3 Database Design

Contract Obligation Tracking & Compliance Management Platform

Objective

This sprint defines the database structure required for ContractIQ before additional backend features are implemented.

users

Central identity table for authentication, roles, departments and account status.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	User identifier
name	VARCHAR(100)		Full name
email	VARCHAR(255)	UNIQUE	Login/contact email
hashed_password	VARCHAR(255)		Password hash
department	VARCHAR(100)		Department
role	VARCHAR(50)		Application role
is_active	BOOLEAN		Account status
created_at	TIMESTAMP		Creation time
updated_at	TIMESTAMP		Last update

contracts

Central business entity storing contract parties, ownership, lifecycle dates, status and value.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Contract identifier
contract_number	VARCHAR(100)	UNIQUE	Business contract number
title	VARCHAR(255)		Contract title
party_name	VARCHAR(255)		External party
owner_id	BIGINT	FK → users.id	Internal owner
status	VARCHAR(50)		Lifecycle status
start_date	DATE		Start date
end_date	DATE		Expiry date
value	DECIMAL(15,2)		Contract value
created_at	TIMESTAMP		Creation time
updated_at	TIMESTAMP		Last update

contract_versions

Preserves historical contract document versions and who created each version.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Version identifier
contract_id	BIGINT	FK → contracts.id	Parent contract
version_number	INT		Version number
document_path	VARCHAR(500)		Document location
change_summary	TEXT		Changes
created_by	BIGINT	FK → users.id	Creator

Column	Data Type	Key / Constraint	Purpose
created_at	TIMESTAMP		Creation time

obligations

Tracks contractual duties, responsible users, deadlines, priorities and completion.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Obligation identifier
contract_id	BIGINT	FK → contracts.id	Parent contract
assigned_to	BIGINT	FK → users.id	Responsible user
description	TEXT		Obligation details
due_date	DATE		Due date
status	VARCHAR(50)		Progress status
priority	VARCHAR(20)		Priority
completed_at	TIMESTAMP		Completion time
created_at	TIMESTAMP		Creation time

renewals

Tracks renewal requests, proposed dates, outcomes and notes.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Renewal identifier
contract_id	BIGINT	FK → contracts.id	Contract
requested_by	BIGINT	FK → users.id	Requester
renewal_date	DATE		Proposed renewal date
new_end_date	DATE		New expiry
status	VARCHAR(50)		Renewal status
notes	TEXT		Notes
created_at	TIMESTAMP		Creation time

notifications

Stores alerts for users, including deadlines, overdue items and renewals.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Notification identifier
user_id	BIGINT	FK → users.id	Recipient
contract_id	BIGINT	FK → contracts.id	Related contract, optional
obligation_id	BIGINT	FK → obligations.id	Related obligation, optional
type	VARCHAR(50)		Notification type
title	VARCHAR(255)		Title
message	TEXT		Content
is_read	BOOLEAN		Read status
created_at	TIMESTAMP		Creation time

reports

Stores generated report metadata, filters and file locations.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Report identifier
generated_by	BIGINT	FK → users.id	Generator
report_type	VARCHAR(100)		Report type
title	VARCHAR(255)		Title
parameters	JSONB		Report filters
file_path	VARCHAR(500)		File location
created_at	TIMESTAMP		Generation time

audit_logs

Records important actions for accountability, including affected entity and old/new values.

Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Audit identifier
user_id	BIGINT	FK → users.id	Actor
entity_type	VARCHAR(100)		Affected entity
entity_id	BIGINT		Affected record
action	VARCHAR(50)		Action
old_values	JSONB		Previous values
new_values	JSONB		New values
ip_address	VARCHAR(45)		IP address
created_at	TIMESTAMP		Audit time

activities

Stores the operational timeline of comments, reviews, assignments and other activities.

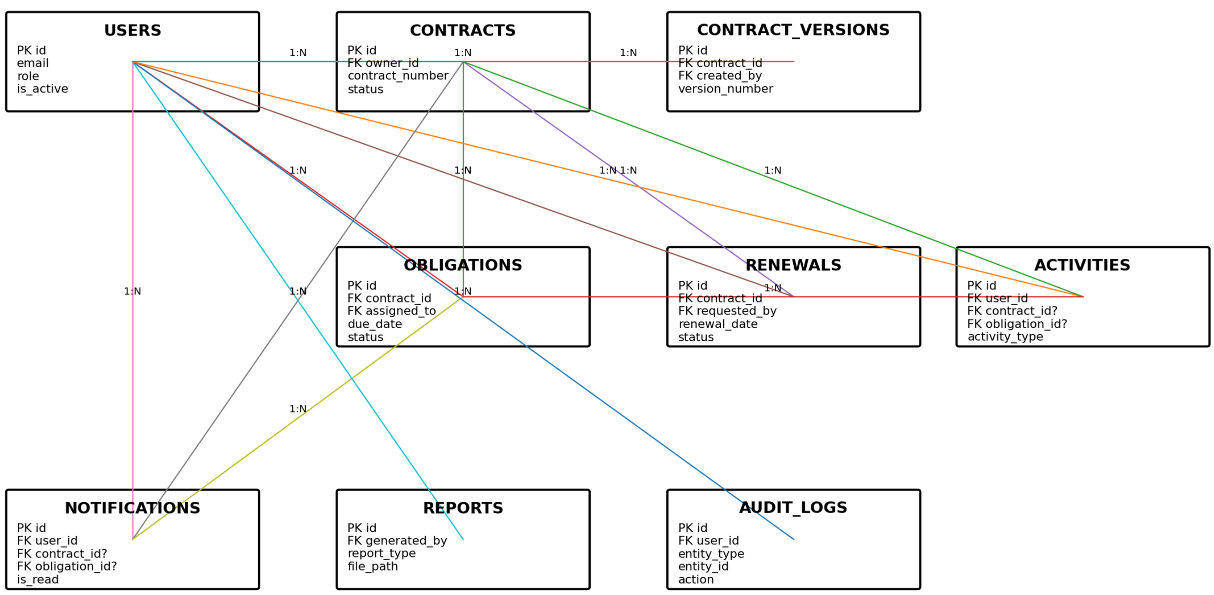
Column	Data Type	Key / Constraint	Purpose
id	BIGINT	PK	Activity identifier
user_id	BIGINT	FK → users.id	Actor
contract_id	BIGINT	FK → contracts.id	Related contract, optional
obligation_id	BIGINT	FK → obligations.id	Related obligation, optional
activity_type	VARCHAR(100)		Activity type
description	TEXT		Details
created_at	TIMESTAMP		Activity time

Table Relationships

Parent	Child	Relationship	Foreign Key
users	contracts	1:N	contracts.owner_id → users.id
contracts	contract_versions	1:N	contract_versions.contract_id → contracts.id
users	contract_versions	1:N	contract_versions.created_by → users.id
contracts	obligations	1:N	obligations.contract_id → contracts.id
users	obligations	1:N	obligations.assigned_to → users.id
contracts	renewals	1:N	renewals.contract_id → contracts.id
users	renewals	1:N	renewals.requested_by → users.id
users	notifications	1:N	notifications.user_id → users.id
contracts	notifications	1:N optional	notifications.contract_id → contracts.id
obligations	notifications	1:N optional	notifications.obligation_id → obligations.id
users	reports	1:N	reports.generated_by → users.id
users	audit_logs	1:N	audit_logs.user_id → users.id
users	activities	1:N	activities.user_id → users.id
contracts	activities	1:N optional	activities.contract_id → contracts.id
obligations	activities	1:N optional	activities.obligation_id → obligations.id

Entity Relationship Diagram

ContractIQ – Sprint 3 Entity Relationship Diagram



How the Tables Work Together

Users provide identity and role information. Contracts are the central business records; versions preserve document history and obligations store duties and deadlines. Renewals manage contract extensions. Notifications provide alerts, reports store generated-report metadata, activities provide the operational timeline, and audit logs provide accountability.

The main relationships are one-to-many. Future many-to-many requirements can be supported with junction tables.